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Behaviours, Institutions, and the Architecture of Competitive-Social Networks

*A framework for describing, measuring, and comparing human
networks organized around shared practice*

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Abstract

This paper proposes a framework for describing human networks that form around a shared competitive or creative practice — esports scenes, sports fandoms, hobbyist communities, and adjacent structures. Its central claim is that such a network is better understood as a set of **behaviours people activate into and out of** than as a population divided into fixed segments ("players," "viewers," "fans"). A person who plays today may watch tomorrow and discuss both on the weekend; the same behaviours recur across what look like very different communities, which means they can serve as a common measurement language across networks that would otherwise seem incomparable. The framework defines a small set of **participation behaviours**, a companion concept of **institutions** (the persistent structures sustained behaviour crystallizes into), and a set of **dimensions** — institutionalization, parasocial attachment, social embeddedness — that cut across behaviours rather than belonging to any one of them. It argues that network "types" should be discovered by clustering real networks on these measurements, not assumed in advance, and that claims about which behaviour drives a network's growth should be tested empirically per network rather than built into the framework as an axiom. The paper closes with a discussion of how such a framework might be visualized, and situates the approach relative to existing literature in diffusion theory, parasocial relationship research, communities of practice, and institutional sociology.

1. Why this framework exists

Most existing ways of describing a competitive or creative community rely on a single proxy measurement, chosen because it happens to be the easiest thing to get historical data on, and then ask that proxy to answer a much bigger question than it was built for. A natural example: the age of an organized, prize-bearing competitive circuit around a given practice is often used as a stand-in for "when did this community mature." It's a reasonable instinct — professional competition is visible, dated, and institutionally documented in a way that grassroots participation usually isn't — but it quietly substitutes one narrow, late-arriving signal for a much broader question.

The substitution breaks in a specific, checkable way. A professional competitive tier requires an existing base of participants large enough to produce elite performers and organizations willing to invest in them; it cannot exist before that base does. So a measure built entirely from competitive-tier activity is necessarily a **lagging indicator** — it tells you when the community had *already* matured enough, for long enough, to support professionalization, not when the community itself formed or how big it actually is. Worse, a single scalar measure of this kind can only describe one axis of what is, in practice, a multivariate situation: a community can be large and devoted to watching without ever developing an organized play tier, or it can have a thriving grassroots practice with no spectator culture

to speak of. No single number — however well-measured — can summarize a state that genuinely has more than one dimension.

This paper's starting premise, then, is that the unit of analysis should not be "the competitive scene" treated as one thing, measured one way. It should be the **behaviours** that constitute a network's activity, measured separately and comparably, so that a network's actual shape — not a single borrowed proxy for it — becomes visible.

2. What kind of network does this describe

The framework describes what we call a **social-competitive network**: a group of people whose connection to each other and to a shared practice is constituted by what they *do*, not by a fixed role they occupy. Nobody is permanently "a player" or "a viewer" in this model — those are behaviours a person activates into for a bounded period, which is why the same person can occupy several of them in the same week, or the same evening.

This has a direct consequence for measurement. If behaviours are activated states rather than population segments, a network cannot be adequately described by headcounts of who belongs to which group. It has to be described by **readings on each behaviour, taken across the whole population, over time** — how much Play is happening, at what level of institutional organization, how much Watch, directed at what kind of source, and so on. The network's "shape" is the resulting profile across all of these readings, not a pie chart of member types.

A second consequence: because the same behaviour-vocabulary applies regardless of what the underlying practice actually is — a competitive video game, a traditional sport, a card game, a creative sandbox — networks that look superficially unrelated become genuinely comparable. The framework is deliberately agnostic about the delivery substrate. What matters is whether a persistent, competitive or creative, community practice exists around something, not whether that something runs on particular software.

3. The core architecture

3.1 Participation behaviours

A **participation behaviour** is a voluntary, individually performed act, bounded in time, whose performance is what constitutes a person's membership in the network at that moment. Behaviours are episodic and renewable, not permanent status changes — a person activates into Play today and may not tomorrow.

Working through a wide range of candidate behaviours and repeatedly applying a simple test — *is this a genuinely distinct act with its own output or goal, or is it better modelled as a property that modifies*

some other act — narrows the list to five that survive as independent behaviours:

- **Play** — direct participation in the core practice itself, spanning casual through highest-stakes competing. Institutional context (see §3.2) is tracked separately rather than treated as a different verb.
- **Watch** — spectating, live or recorded.
- **Discuss** — commentary, analysis, and meta-discourse about the practice or its participants.
- **Attend** — synchronized physical co-presence at an event. (An earlier version of this framework also allowed "virtual attendance"; on inspection this collapses entirely into Watch at high social intensity and was removed — see §3.3.)
- **Create** — producing an output that becomes part of the network and would not otherwise exist. This single behaviour covers a wider range than it first appears: broadcasting oneself playing, writing a strategy guide, producing fan art, and organizing a tournament are all instances of the same underlying act — composing a consumable output — differentiated by output type rather than by separate verbs.

Several other candidates were considered and did not survive as independent behaviours. "Elite play" and "professionalized play" are not separate acts from Play — they are the same act read at different points on the institutionalization dimension. "Streaming" is not a Play-variant but a Create-instance (it composes a broadcast output). "Organizing" an event dissolves into Create (producing an event-as-output) and Discuss (community moderation). Two further candidates — Purchase and Socialise — looked at first like behaviours and turned out, on closer inspection, to be something else; both are discussed in §3.4.

A behaviour's readings co-occur rather than compete. A person attending a live event while competing is simultaneously doing Attend and Play; a person streaming themselves playing is simultaneously doing Play and Create. The framework does not ask "which behaviour is this person doing" as though the categories were mutually exclusive — it asks "what is this person's reading on each behaviour, right now," and several can be non-zero at once.

3.2 Institutions

An **institution** is a persistent, supra-individual, nameable structure — a league, a company, a team, a recognized tournament series, a community body — that forms when sustained participation behaviour crosses a threshold sufficient to justify an outside party (a publisher, a sponsor, an organizer, a platform) investing in formalizing it. Institutions outlive any single participant and any single instance of the behaviour that produced them. They have their own lifecycle — formation, growth, stabilization, and eventual decline — and, critically, they feed back and reshape the future rate and shape of the very behaviour that produced them. A franchised, professionally organized competitive league, for instance, does not merely reflect an already-large player base; once it exists, it changes how players move between teams, how long careers last, and who enters the practice in the first place. Institutions are not passive residue. They are active participants in a network's subsequent evolution.

Institution is not a second category sitting beside behaviours in a parallel list. It is the **reified, named state a behaviour's own institutionalization reading produces once that reading crosses a threshold and holds**. Any measurement scheme that already tracks "when did organized competitive activity around this practice first reach a sustained, multi-region scale" is, in effect, already doing this — dating the formation of one specific institution — without necessarily recognizing it as one instance of a general mechanism that applies to every behaviour, not just Play.

3.3 Dimensions that cut across behaviours

Two families of dimension apply across multiple behaviours rather than belonging to any single one.

Institutionalization (behaviour-side) measures how organized and formalized the context of a behaviour-instance is, ranging from none through informal, semi-organized, to fully institutional. It applies natively to Play. It also applies to Watch, Discuss, and Attend — but there it measures the institutionalization of *what is being consumed or attended*, not of the consuming act itself. Watching an individual creator's own broadcast and watching an officially produced tournament broadcast are structurally different experiences even at identical viewing volume, and collapsing that distinction into a single "watch hours" figure — or worse, into a presumed category label for the title being watched — hides exactly the thing that often matters most.

Parasocial attachment and social embeddedness (person-side) are kept in the framework deliberately even though neither is currently measurable from the kind of trace data (viewership logs, transaction records, participation counts) that is typically available. The reasoning for naming an axis that cannot yet be measured is straightforward: a named, unmeasured dimension can be picked up the moment a measurement technique becomes available; a dimension that was never named cannot be retrofitted later without first re-deriving why it matters.

Parasocial attachment is a one-directional relationship between a person and the object of their attachment — who does not know the person exists. Social embeddedness is, by contrast, the degree to which a behaviour is co-experienced with a person's own real, reciprocal relationships. These are not opposite ends of one spectrum, despite an intuitive pull toward treating them that way. They are **orthogonal**: a person can watch a favourite performer deeply attached and alone, or watch the same performer as a genuinely shared social occasion with their own friend group, or attend a large shared event mainly for the communal occasion with no strong attachment to any specific participant at all. Four real, distinguishable combinations exist, not two.

A further refinement worth recording: a parasocial attachment-object need not be an individual. Devotion to an institution — a team, a brand — is still structurally one-directional, even though the object is an organization rather than a person.

3.4 Purchase and Socialise: two behaviours that turned out to be something else

Two further candidates illustrate the value of applying the same collapse test rigorously rather than accepting a familiar-sounding verb at face value.

Purchase looks, at first glance, like an obvious behaviour: people spend money in relation to the network. But examined closely, a purchase is almost always better understood as a *quantified expression* of something else — paid access to another behaviour (a subscription is paid Watch; an entry fee is paid Play), or a monetized signal on the parasocial or social-embeddedness axes (a direct donation to a specific performer is parasocial expression; buying branded apparel to display community membership to one's own social circle is social expression). Spend amount itself is also a less stable unit than it first appears, since purchasing power varies substantially across regions; a comparative framework is better served by counting discrete purchase decisions than by summing currency value across contexts. Purchase survives in the framework as a measurement channel that can be read off several other axes, not as its own behaviour.

Socialise looked, initially, like a natural companion to Discuss — another social verb alongside spectating and talking. It does not survive scrutiny as such, for a structural reason: Discuss is genuinely episodic, a bounded verbal exchange, the same shape as the other four surviving behaviours. Socialise is not. "Becoming socialised within a network" reads more like a state a person enters and holds — private, then embedded — than like a discrete, repeatable act. Pressed further, this turns out to be the person-level mirror image of institutionalization itself. Institutionalization aggregates one behaviour across an entire population to produce a network-level object: an institution. The state a person moves into through sustained, high-intensity, socially embedded participation is the same operation performed the other way — aggregating *all* of one person's behaviours, across time, to produce an individual-level object. We refer to this provisionally as **centrality**, or equivalently **investment**: a person's cumulative position in the network, driven jointly by overall behavioural intensity and by social embeddedness specifically. Like institutions, this state plausibly feeds back into future behaviour — a highly invested member should be more resistant to leaving through a quality dip in the practice, more likely to recruit others, and more likely to defend the community publicly — though this remains a hypothesis carried by the structural parallel rather than something directly demonstrated.

4. Modes should be discovered, not assumed

It is tempting, once a vocabulary of behaviours exists, to sort networks into a small number of named types — a competitive-practice type, a narrative-consumption type, a social-sandbox type — and then describe each type's characteristic behaviour profile. This framework deliberately resists doing that as a first step. The number and shape of network "types," if they exist at all, should be an **output** of measuring many real networks on the behaviour and institution vectors described above and clustering them, not an **input** chosen in advance and then confirmed by selectively noticing supporting evidence.

A cluster found this way should additionally be tested against a null model — compared to what the same clustering procedure would find in randomly rewired data — so that apparent types can be distinguished from patterns that would appear in unstructured data by chance. Named categories are a legitimate and useful endpoint; they are not a legitimate starting assumption.

5. Precedence is an analytical output, not a taxonomic commitment

A separate temptation is to decide, as part of defining the framework, which behaviour is the "prime mover" that drives the others — the intuitive candidate being Play, on the reasoning that one must participate before one can watch, discuss, or attend. This intuition is worth taking seriously but not worth encoding as an axiom. Watching, in particular, has real channels that do not require a pre-existing base of players at all — algorithmic discovery, a single popular broadcaster, spectacle appeal independent of participation — and a practice can plausibly build a substantial audience around high-level performance *before* that performance becomes institutionally organized, not after.

The resolution this paper proposes is to keep the behaviour taxonomy flat and symmetric — no behaviour privileged by definition — and to treat "which behaviour leads" as a **per-network empirical question**, answered by testing lead-lag relationships across behaviour time series once those time series exist. The claim "participation drives everything else" then becomes a testable, falsifiable statement — true for some networks, false or reversed for others — rather than a premise baked into the measurement instrument itself. A framework whose own structure already assumes the answer cannot be used to discover that the answer sometimes differs.

6. What this framework is designed to enable

Four capabilities motivate the design choices above:

Cross-network comparability without presupposed categories. Because the same behaviour vocabulary applies regardless of the underlying practice, two networks that look nothing alike on the surface can be compared directly on measured axes, rather than compared only within an assumed category.

Emergent typing. Network "types," to the extent they are real, should fall out of clustering real measurements — and be validated against a null model — rather than being assumed and then illustrated.

Precedence and causal analysis. Keeping behaviours as separable, comparably measured quantities makes it possible to ask genuine ordering and causal questions between them — does watching tend to precede or follow institutionalized play, and does the answer vary by practice — questions that collapse into unanswerable the moment behaviours are folded into one another.

Honest accounting of what cannot (yet) be measured. Parasocial attachment and social embeddedness are retained in the framework explicitly, with no proxy forced onto them, so that any analysis built on this framework can state plainly which dimensions it could actually observe and which it could only name. As measurement techniques mature — richer platform data, new instruments, new consented data sources — previously unmeasurable axes can be picked up without having to first rediscover that they matter.

7. Visualization: a research direction, not yet built

A framework built from a small number of comparable, numeric axes per network invites a direct visual treatment, and it is worth sketching what that could look like, even though building it is a separate undertaking from the conceptual work above.

The basic shape. Each of the core behaviours (Play, Watch, Discuss, Attend, Create) can be drawn as a spoke radiating from a central point representing the network itself — the same underlying idea as a radar or spider chart, extended into three dimensions. Spoke length encodes behavioural intensity; a second visual channel along each spoke — colour, texture, or a layered shell around the spoke's tip — encodes that behaviour's institutionalization reading. The resulting polyhedral shape becomes a network's visual signature: two networks with genuinely different behaviour profiles produce visibly different solids, and because every network is drawn against the *same* fixed set of spokes, shapes are directly comparable by overlaying them, which is precisely the design goal the underlying measurements were built to support.

Representing what cannot be measured. Dimensions this project cannot currently observe — parasocial attachment, social embeddedness, most of Purchase — should not simply be omitted from the picture. Rendered as translucent, dashed, or desaturated spokes at an explicitly marked "unknown" length, they keep the dimension visually present without asserting a value that was never measured. A viewer should be able to tell, at a glance, which parts of a network's shape are real readings and which are acknowledged gaps.

Population distribution, not just network aggregate. Centrality/investment (§3.4) is a per-person quantity, not a per-network one, and could be layered onto the same visualization as a field of points scattered around the solid — points near the core representing highly invested individuals, points toward the periphery representing more occasional participants — turning the shape from a single aggregate summary into a visible distribution of where real people actually sit within it.

Time as a dimension. Because networks evolve, the same shape could animate: a title's polyhedron morphing year by year as institutionalization rises on some spokes and intensity shifts on others, with a simple scrubber allowing a viewer to move through a network's history and watch its signature change shape — a direct, intuitive rendering of the adoption-curve question this framework was built to investigate in the first place.

A longer-term vision. None of the above requires novel technology — standard web-based 3D rendering (WebGL-class tooling) and interactive charting libraries are more than sufficient, and the harder work is in the underlying measurement pipeline, not the rendering layer. The natural long-term endpoint is an interactive site where any tracked network's shape can be explored, compared side by side with others, and scrubbed through time, with every greyed-out spoke serving as a standing, visible invitation to go measure the thing that spoke represents.

8. Related literature

This framework was developed independently, from applied work on digital fandom and competitive-practice networks, rather than as a direct application of any single existing theory. Several points of connection are worth recording, both to situate the work and because each connection suggests a direction the framework could be deepened.

The behaviour-over-population-segment starting premise echoes **uses and gratifications** research in media studies (Katz, Blumler & Gurevitch), which asks what functions media consumption serves for an individual rather than treating an audience as an undifferentiated mass. The parasocial attachment dimension traces directly to **Horton and Wohl's** foundational 1956 account of parasocial interaction — the one-directional relationship a person can form with a media performer who does not know they exist — a concept that has since been extended, by a growing literature, specifically into livestreaming contexts. The institutionalization dimension and the account of how institutions form and subsequently reshape the behaviour that produced them connects to **institutional theory** in organizational sociology, particularly DiMaggio and Powell's account of how fields professionalize and converge once a critical mass of activity exists to justify it. The adoption-curve framing in §1 and §5 draws on the classical **diffusion of innovation** literature (Rogers) and the **Bass diffusion model**, which separates adoption into external influence (marketing, media exposure) and internal influence (word of mouth, imitation) — a distinction plausibly relevant to why Watch and Play can decouple, since the two behaviours may be driven by different mixes of these two forces. The competitive-practice behaviour itself — persistent, community-organized, skill-based practice, independent of its delivery substrate — sits close to **communities of practice** as developed by Lave and Wenger, and the Create behaviour's broader treatment of fan-produced output connects to the **participatory culture** tradition in fandom studies associated with Henry Jenkins. Finally, the professionalization of competitive gaming specifically has its own dedicated academic treatment, notably T.L. Taylor's work on the rise of organized esports, which engages directly with the institutionalization question this framework treats as one axis among several rather than the central phenomenon.

None of these connections were the starting point for the framework; each was recognized after the fact, as the framework's own internal logic arrived at a position that existing literature had already mapped from a different direction. That convergence is worth taking as mild evidence the framework is tracking something real, not as a claim that it was built on any of this literature directly.

9. Open questions

This remains a working framework, not a finished one. Unresolved items, in roughly descending order of how foundational they are:

- Whether Attend survives as an independent behaviour or is better modelled as a physical-co-presence context that modifies Play, Watch, or Discuss.
- Whether centrality/investment is correctly named, and whether its inputs are limited to behavioural intensity and social embeddedness or whether parasocial attachment also contributes.
- Whether institutionalization is a genuinely independent axis for Purchase or is redundant with the institutionalization of whichever behaviour a given purchase is attached to.
- Whether adversarial structure (competitive, cooperative, or solo) and skill level need to be tracked as their own dimensions on Play, separate from institutionalization.
- Drafting the behaviour and institution tables for network types beyond competitive-practice — narrative-consumption and social-creative-sandbox candidates in particular — which is likely a precondition for the emergent-clustering approach in §4 to have more than one mode's worth of data to work with.
- Building the per-network, per-behaviour-pair lead-lag analysis described in §5, which requires assembling time series per behaviour that do not yet exist in usable form.
- The visualization direction in §7, which is a substantial undertaking in its own right and has not been started.